Comparing RGB Color Features and YOLO-Based Deep Learning for Noninvasive Anemia Detection from Conjunctival Images

1st Pranav Sachdeva
Oakton High School
Vienna, USA
pranavsachdeva1109@gmail.com

2nd Shashank Edupuganti
Westfield High School
Chantilly, USA
shashankedupuganti@gmail.com

3rd Abhinav Karthik
Westfield High School
Chantilly, USA
reachabhi.ak@gmail.com

4th Andreea Sistrunk
Technical Education and Workforce Development
Virginia Tech
Alexandria, USA
sistrunk@vt.edu

***Abstract*—Anemia is a common condition that is traditionally diagnosed through blood testing, which can be invasive, time-consuming, and resource-intensive. This study investigates a noninvasive approach for anemia screening using conjunctival images, comparing two methods: a classical color-based approach using RGB and HSV color statistics, and a deep learning approach based on YOLOv8 classification. A publicly available dataset of conjunctival images was cleaned, deduplicated, and screened for synthetic (SMOTE-generated) images to reduce leakage and improve reproducibility, yielding 157 unique real images. Both approaches were evaluated under identical five-fold stratified cross-validation splits, allowing a direct, matched comparison. The color-based method leverages the observation that conjunctival pallor is reflected in changes in color intensity and variability, while the YOLOv8 model learns image features directly from the eye region; Grad-CAM++ visualizations are used to check which regions drive the deep model's predictions. External validation on images collected independently by the research team has not yet been performed and is described here as planned future work. Paired McNemar testing finds no significant difference in overall accuracy between the two approaches, and their areas under the ROC curve are statistically indistinguishable, but the deep model holds a significant advantage in sensitivity. The two therefore differ less in how well they rank images than in where their decision thresholds fall – a distinction that matters directly for screening. Results from this study clarify whether a simpler, interpretable color-based baseline can approach the performance of a learned vision model, and whether noninvasive anemia screening from conjunctival images is feasible in low-resource or early-screening settings.**

***Index Terms*—anemia screening, conjunctival images, noninvasive diagnosis, YOLOv8, logistic regression, medical image classification, cross-validation, Grad-CAM++**

I. Introduction

Anemia remains one of the most prevalent global health conditions, affecting an estimated 1.92 billion people worldwide as of 2021, a global prevalence of 24.3% across all ages [1]. Diagnosis conventionally requires a blood draw and laboratory analysis of hemoglobin concentration: accurate, but invasive, resource-intensive, and poorly suited to large-scale or low-resource screening. This has led to a substantial increase in research interest in non-invasive conjunctival anemia screening. The palpebral conjunctiva visibly pales with reduced hemoglobin concentration and, unlike skin, is largely unaffected by pigmentation differences across populations [2], [3].

Several public datasets have supported this line of research. Dimauro et al. introduced the Eyes-Defy-Anemia dataset — 218 conjunctiva images from Italian and Indian patients paired with laboratory hemoglobin measurements — and demonstrated anemia classification using engineered color features with a RUSBoost classifier to address class imbalance [4], [5]. Separately, Appiahene et al. introduced the larger CP-AnemiC dataset (710 images from Ghanaian children) alongside a joint classification-and-regression deep network [6], and a related smartphone application backed by a FastAPI server [7] — a system architecture notably similar in structure to the one used in this work. More recent work has applied larger convolutional and object-detection architectures to this task: an "improved" YOLOv9 variant reporting near-ceiling metrics (F1 = 1.00, mAP@50 = 0.995) on a heavily augmented custom dataset [8]; vision transformer approaches with attention-based explainability [9]; a stacking ensemble of VGG16, ResNet-50, and InceptionV3 on GAN-augmented conjunctiva images, reporting an AUC of 0.97 [10]; and a separate multi-architecture approach combining AlexNet, DenseNet, and VGGNet for infants aged 6–59 months, reporting 98.79% accuracy [11].

These prior systems demonstrate the promise of deep learning for conjunctiva-based anemia screening, but several report near-ceiling performance without disclosing cross-validation methodology, deduplication, or leakage-prevention procedures, and none compare a deep architecture against a simpler interpretable baseline under matched conditions. This work addresses that gap: a YOLOv8 classifier trained via five-fold stratified cross-validation on a deduplicated dataset is compared directly against a logistic-regression baseline on

979-8-3195-2794-3/26/$31.00 © 2026 IEEE

color and texture features, under identical splits, allowing a fair assessment of what the deep model learns beyond raw conjunctival color. In preparing the dataset we also identified and excluded synthetic images generated by SMOTE oversampling that had been mixed into the source data – a data-quality issue that merits attention given how often small, imbalanced medical-imaging datasets are extended with oversampling techniques not designed for image data.

The working dataset that survives this cleaning is deliberately small: 157 unique real images, down from 288 files. We treat this as a property of the study design rather than a shortcoming. Small, imperfect image collections are the normal starting condition in the low-resource settings where non-invasive screening would matter most, and inflating such a dataset through duplication or synthetic oversampling raises reported numbers without improving the model. This paper is framed as a proof of concept for what is achievable under realistic data constraints, using procedures intended to transfer to other small-dataset medical imaging problems.

II. Methods

A. Data Collection

The dataset used in this study was obtained from a publicly available conjunctiva image collection hosted on Roboflow Universe [12], released under a CC BY 4.0 license. The collection consists of images labeled as either anemic or normal. Because the dataset was previously collected and did not include corresponding laboratory hemoglobin measurements, all analyses in this study rely on the properties and labels of the images themselves rather than on clinically measured hemoglobin levels.

B. Data Processing

Initial inspection of the local copy of the dataset (288 image files across the original train and test folders) revealed several data-quality issues common to small, re-hosted medical-imaging datasets: duplicate and rotated copies of the same underlying image saved under different filenames, images duplicated across the original train and test folders (a data-leakage risk in the original, since-abandoned split), and synthetic images generated via SMOTE oversampling that had been mixed directly into the anemia class.

To address these issues, every image was hashed using a rotation-invariant perceptual hash (`imagehash.phash`, computed across 0°, 90°, 180°, and 270° rotations and keeping the minimum hash as a canonical identifier), which collapses duplicate and rotated copies of the same source image into a single representative image regardless of orientation or filename. Images with "smote" present in their filename were identified and set aside separately, since SMOTE performs pixel-level interpolation between feature vectors and is not designed for raw image data; these synthetic images are visually distinguishable by an unnatural, purplish blending artifact and were excluded from all reported results. After deduplication and SMOTE removal, the working dataset consisted of 157 unique, real images: 69 labeled anemia and 88 labeled normal. Each retained image was assigned a clean sequential identifier, and a manifest file was generated recording the mapping from each identifier to its original filename, class label, and eventual cross-validation fold assignment, to support full reproducibility of the data pipeline. Fig. 1 summarizes this pipeline end to end.

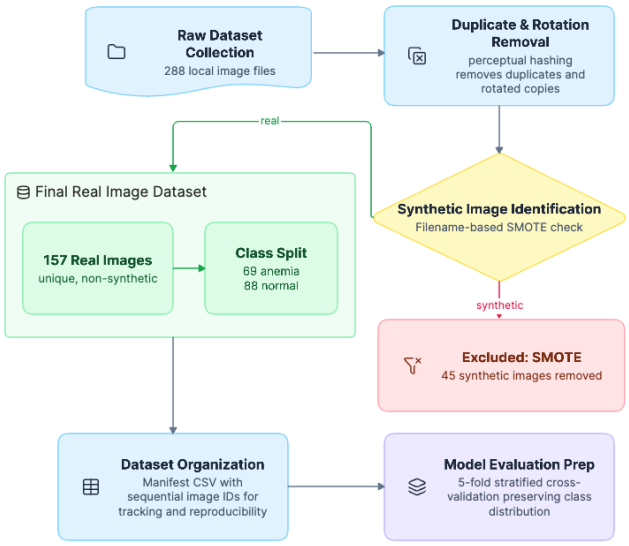


Fig. 1. Data cleaning pipeline: from the raw local dataset to the final 157-image, fold-assigned dataset.

C. Cross Validation

Given the limited dataset, and particularly the 69 anemia images, a single fixed split was judged too unstable for reliable reporting: a conventional 70/15/15 split would leave roughly 11 anemia images in the test set, making metrics highly sensitive to which images happened to fall there. Five-fold stratified cross-validation was used instead (scikit-learn `StratifiedKFold`, `random_state=42`), preserving class balance within each fold. Results for both approaches are reported as the mean $\pm$ standard deviation across folds, using identical fold assignments to allow a direct, matched comparison.

D. YOLOv8 Classification Model

A convolutional neural network classifier was implemented using the YOLOv8 classification framework (Ultralytics 8.4.96) [13]. Rather than training a network from randomly initialized weights, the model was initialized from a YOLOv8n classification backbone pretrained on ImageNet and fine-tuned on the conjunctiva dataset via transfer learning, a standard approach for small medical-imaging datasets where training a deep network from scratch is prone to severe overfitting. A separate model was trained independently on each of the five cross-validation folds. Each was fine-tuned from the YOLOv8n-cls backbone for a maximum of 100 epochs at 640×640 input resolution with a fixed random seed of 42, using early stopping with a patience of 20 epochs to halt

training once validation performance stopped improving. In practice the best-performing epoch on every fold was found well before the 100-epoch maximum, indicating that the model converges quickly on this dataset and then plateaus.

To provide qualitative interpretability, Gradient-weighted Class Activation Mapping++ (Grad-CAM++) [14] was applied to a convolutional block of the trained network, producing a heatmap of the regions most influencing the predicted class. The target layer was chosen empirically. The final convolutional block yields a 20×20 activation grid at 640×640 input, so each cell covers 32×32 input pixels – wide enough to span conjunctival tissue and adjacent background at once. Inspecting the grid before it is upsampled for display confirmed that this overlap, rather than the smoothing applied when the heatmap is drawn, is what produces the apparent spillover. An earlier block with a 40×40 grid (16×16 pixels per cell) tracked the conjunctival boundary substantially more tightly and was used for the final figure. Fig. 2 shows overlays from that layer, generated with the fold-0 model on its own held-out split and arranged as a confusion matrix; the false-negative panel is the only anemia image this fold misclassified. Activations concentrate along the conjunctival tissue and its margins in all four cases, with some residual activation on the eyelid margin and adjacent background consistent with the coarse activation grid described above. The overlays indicate that the model's attention is tied to the conjunctiva rather than to background cues, though this remains a qualitative check rather than proof.

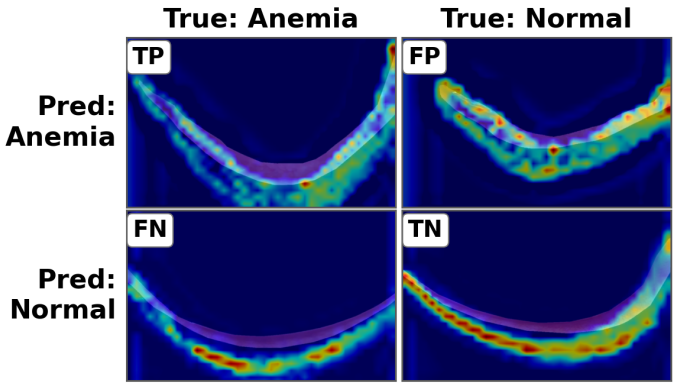


Fig. 2. Grad-CAM++ activation heatmaps from the fold-0 cross-validation model, arranged as a confusion matrix: columns give the true label, rows the predicted label. Panels are cropped vertically to the band containing all activation.

E. RGB/HSV Logistic Regression Baseline

To provide a classical machine learning baseline, a logistic regression classifier was trained using handcrafted color features extracted from each image.

For every image, twelve statistical features were computed: the mean and the standard deviation of each of the three RGB channels (R, G, B) and of the three HSV channels (H, S, V). These features summarize both the average color and the variability of color within the conjunctiva, allowing the baseline model to capture differences in both pigmentation and local texture.

Prior to model training, feature standardization was performed by fitting the scaler only on the training portion of each fold and applying the learned transformation to the corresponding validation data. This prevented information leakage between the training and validation sets.

The logistic regression model was trained and evaluated using the same five-fold cross-validation protocol as the YOLOv8 classifier to ensure a direct comparison between the two approaches.

To provide an interpretability comparison alongside the YOLOv8 classifier's Grad-CAM++ visualizations, a final logistic regression model was fit on the full 157-image dataset (combining the training and validation portions of one cross-validation fold, which together cover all unique images) and inspected directly via its standardized coefficients. All twelve coefficients are shown in Fig. 3, ordered by magnitude and signed so that positive values push the prediction toward anemia. Blue-channel variability was the strongest positive predictor of anemia, with red-, value-, and hue-channel variability the strongest negative predictors. This suggests that, beyond simple average color, the local color variability of the conjunctiva – plausibly reflecting the smoother, less textured appearance of pale tissue relative to well-perfused tissue – carries meaningful signal for this classical baseline, offering a concrete, interpretable counterpart to the YOLOv8 model's learned representation.

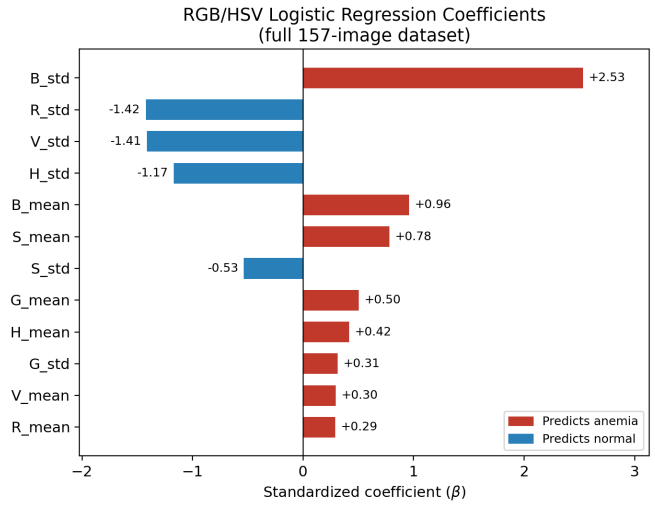


Fig. 3. Standardized logistic regression coefficients for all twelve RGB/HSV features, fit on the full 157-image dataset. Positive coefficients push the prediction toward anemia; negative coefficients push it toward normal.

F. Statistical Analysis

Because both models were evaluated on identical cross-validation splits, every image carries a prediction from each approach, permitting a paired rather than an independent comparison. Predictions from the five fold models were pooled to give 157 held-out decisions per approach. These come

from five separately trained models rather than a single model evaluated once, which is standard practice for cross-validated reporting.

Agreement was assessed with McNemar's test in its exact binomial form, the chi-square approximation being unreliable at small discordant counts. Three paired comparisons were made – overall accuracy across all 157 images, sensitivity on the 69 anemia images, and specificity on the 88 normal images – so a Bonferroni-adjusted threshold of $\alpha = 0.0167$ applies. Pooled estimates carry two-sided 95% Wilson score confidence intervals. Because McNemar evaluates hard predictions at each model's default threshold, threshold-independent discrimination was also compared by area under the ROC curve (AUC), computed per fold and compared with a paired t-test. The five fold checkpoints and all per-image predictions were retained, and training reproduces exactly under the fixed seed.

G. App and Distribution System

A mobile application provides an accessible interface for anemia risk assessment. Users capture or upload a conjunctival image using an on-screen guidance box, crop it on-device to isolate the conjunctiva, and choose whether the anonymized image may be retained for future model improvement; images are stored only where consent is given. The cropped image is sent over an encrypted HTTPS connection to a FastAPI backend, which resizes it to 640×640 and runs both the YOLO and RGB/HSV models. Both perform binary classification only, with no severity grading or confidence score returned, and the predicted class is returned as JSON for display in the app.

Clients were built for iOS and Android in Xcode and Android Studio respectively, with a prototype web interface used during backend development. Testing was carried out on a Google Pixel 7 (Android) and an iPhone 15 Pro Max (iOS); end-to-end time from submission to displayed result was under five seconds on both devices. Separating the client from the inference backend allows the model to be updated without changing the application and lets the inference service scale independently.

III. Results

A. YOLOv8-cls Cross-Validation Performance

The YOLOv8n classification model achieved a mean top-1 accuracy of 84.74% $\pm$ 2.20% across the five cross-validation folds. Per-class precision, recall, and F1 are reported in Table I. Early stopping halted training between epoch 26 and 60 on every fold, with the best-performing checkpoint typically found much earlier (epoch 6–12 for four of the five folds), indicating that the model converges quickly on this dataset and then plateaus.

B. RGB/HSV Baseline Cross-Validation Performance

The logistic regression baseline, trained on the same five cross-validation splits, achieved a mean accuracy of 77.78% $\pm$ 6.72%. Per-class precision, recall, and F1 are reported alongside the YOLOv8-cls results in Table I. Following clinical screening convention, anemia is treated as the positive class throughout, so recall on the anemia class is reported as sensitivity and recall on the normal class as specificity. All values in Table I are means $\pm$ standard deviations across the five folds.

TABLE I
YOLOv8-cls vs. RGB/HSV Baseline (5-fold CV, mean $\pm$ std, %)

Metric	YOLOv8-cls	RGB/HSV
Accuracy	84.74 $\pm$ 2.20	77.78 $\pm$ 6.72
Anemia precision	80.02 $\pm$ 5.84	80.45 $\pm$ 7.64
Sensitivity (anemia recall)	88.57 $\pm$ 9.69	65.05 $\pm$ 14.31
Anemia F1	83.44 $\pm$ 2.99	71.20 $\pm$ 10.24
Normal precision	90.83 $\pm$ 6.71	77.03 $\pm$ 7.37
Specificity (normal recall)	81.96 $\pm$ 6.33	87.52 $\pm$ 5.40
Normal F1	85.74 $\pm$ 1.95	81.72 $\pm$ 5.03

C. Statistical Comparison

Pooled across the five held-out folds, the two models agreed on 120 of the 157 images (76.4%) and disagreed on 37. Of those 37 disagreements, YOLOv8-cls was correct on 24 and the RGB/HSV baseline on 13. McNemar's exact test on this margin gives $p = 0.099$, so the difference in overall accuracy is not statistically significant at either the conventional or the adjusted threshold.

Restricting the same paired test to the 69 anemia images gives a different result. The two models disagreed on 20 of them, with YOLOv8-cls correct on 18 and the baseline on 2 ($p = 0.0004$), which remains significant after Bonferroni adjustment. On the 88 normal images the direction reverses: of 17 disagreements the baseline was correct on 11 against YOLOv8-cls's 6, though this difference is not significant ($p = 0.33$).

Ranking ability was comparable: mean per-fold AUC was 0.887 $\pm$ 0.038 for YOLOv8-cls and 0.869 $\pm$ 0.060 for the baseline, a difference of $+0.018$ ($p = 0.71$).

Pooled over all 157 predictions, YOLOv8-cls reached 84.71% accuracy [78.3–89.5], 88.41% sensitivity [78.8–94.0] and 81.82% specificity [72.5–88.5]; the baseline reached 77.71% [70.6–83.5], 65.22% [53.4–75.4] and 87.50% [79.0–92.9] (Wilson 95% intervals). Because the folds are of unequal size (32, 32, 31, 31, and 31 images), these pooled figures differ slightly from the fold-averaged values in Table I; they are the same predictions counted a different way. The sensitivity intervals do not overlap; the accuracy and specificity intervals do.

The code, fold assignments, per-image predictions and trained fold checkpoints underlying every result reported here are available at https://github.com/PranavS1109/conjunctival-anemia-screening.

IV. Discussion

A. Performance Comparison

The two approaches are not statistically distinguishable on overall accuracy. YOLOv8-cls was correct on 133 of 157

held-out images against the baseline's 122, but the models agreed on 120 of them, and the 37 disagreements split 24 to 13 – a margin McNemar's exact test cannot separate from chance ($p = 0.099$). Reporting the roughly 7-percentage-point accuracy gap without that qualification would overstate what 157 images can support, even though YOLOv8-cls did show substantially lower fold-to-fold variance (2.20% vs. 6.72%).

What does distinguish the two models is the kind of error each makes. YOLOv8-cls showed higher recall than precision on the anemia class (88.57% vs. 80.02%), meaning it more often over-called anemia in normal images. The RGB/HSV baseline showed the reverse pattern, with higher precision than recall (80.45% vs. 65.05%), meaning it more often missed true anemia cases. For a screening tool intended to flag individuals for further clinical evaluation, under-calling (a false negative) is the more consequential failure mode, since a missed case receives no follow-up at all, whereas a false positive simply results in an unnecessary but low-cost confirmatory test.

Pooling the 157 held-out predictions makes the practical size of this difference concrete: of the 69 anemia images, the baseline failed to flag 24 against YOLOv8-cls's 8, while on the normal class the two are far closer (11 false positives against 16). Unlike the accuracy gap, this asymmetry is statistically robust: of the 20 anemia images on which the models disagreed, YOLOv8-cls was correct on 18 and the baseline on 2 ($p = 0.0004$, significant after adjustment); the baseline's corresponding advantage on the normal class is smaller and not significant ($p = 0.33$). The comparable AUC values locate this difference at the operating point rather than in discriminative power: a threshold-tuned baseline could reach the same sensitivity, but not for free. Lowering its threshold to match YOLOv8-cls's 88.41% sensitivity drops the baseline's specificity to 67.05% – 29 false positives among 88 normal images against YOLOv8-cls's 16 – a gap averaging 20 points per fold that varies widely and is not significant ($p = 0.15$). The summary is therefore not that one model is better, but that the two trade sensitivity against specificity, and that the YOLOv8-cls reaches the side of that trade a screening tool needs without tuning.

B. Limitations

Several limitations qualify these results, most of them stemming from dataset size. Only 69 anemia images remain after deduplication, limiting statistical power and increasing sensitivity to which particular images were available. The study is underpowered to establish a difference in either overall accuracy or AUC, and both tests confirm this ($p = 0.099$, $p = 0.71$); the class-specific sensitivity result does reach significance but rests on 20 discordant images and should be read as a strong indication rather than a settled effect. Framing the small dataset as a realistic operating constraint does not remove this limitation, only clarifies what the study can support. The dataset also lacks laboratory hemoglobin measurements, so labels reflect the source collection's annotation rather than clinical ground truth, and all conclusions concern agreement with those labels rather than diagnostic accuracy. Finally, no external validation on independently collected images has yet been performed; this is planned future work.

V. Conclusion

This work compared a YOLOv8-cls transfer-learning classifier against an RGB/HSV logistic regression baseline for non-invasive anemia detection from conjunctival images, using five-fold stratified cross-validation on a cleaned, deduplicated 157-image dataset. YOLOv8-cls achieved a mean accuracy of 84.74% $\pm$ 2.20% against the RGB/HSV baseline's 77.78% $\pm$ 6.72%, but a paired McNemar test does not establish this accuracy difference as significant ($p = 0.099$). What the comparison does establish is a difference in error profile: pooled across folds, YOLOv8-cls missed 8 of 69 anemia cases where the baseline missed 24, and on the anemia images where the two disagreed it was correct 18 times to the baseline's 2 ($p = 0.0004$). A mobile application and FastAPI backend were also developed to make this classification accessible outside a research setting.

The more general contribution is the demonstration that a usable screening comparison can be built and honestly evaluated on 157 images. The settings that would benefit most from non-invasive screening are also those least likely to hold thousands of laboratory-confirmed images. A pipeline that treats scarce data as the operating condition rather than an obstacle – deduplicating aggressively, removing synthetic images, holding the evaluation protocol identical across models, and pairing a deep model against an interpretable baseline so its added value is measured rather than assumed – applies well beyond anemia.

Its central claim is correspondingly modest. The results establish that YOLOv8-cls detects significantly more anemia cases than the color baseline at its default operating point, which is the property a screening tool needs. They do not establish an overall accuracy advantage, a difference in ranking ability, or diagnostic accuracy against clinical ground truth. As future work, the team plans to collect and describe a separate, independently gathered set of conjunctival images to externally validate these models beyond the single public dataset used here, which is necessary before any claim of real-world diagnostic performance can be made.

Acknowledgment

The authors thank the faculty of the Virginia Tech Institute for Advanced Computing, and in particular Dr. Jim Egenrieder, for their guidance and support throughout this project.

References

[1] W. M. Gardner, C. Razo, T. A. McHugh, H. Hagins, V. M. Vilchis-Tella, C. Hennessy, H. J. Taylor, N. Perumal, K. Fuller, K. M. Cercy *et al.*, "Prevalence, years lived with disability, and trends in anaemia burden by severity and cause, 1990–2021: findings from the global burden of disease study 2021," *The Lancet Haematology*, vol. 10, no. 9, pp. e713–e734, 2023.

[2] G. Dimauro, D. Caivano, and F. Girardi, "A new method and a non-invasive device to estimate anemia based on digital images of the conjunctiva," *Ieee Access*, vol. 6, pp. 46 968–46 975, 2018.

[3] S. Collings, O. Thompson, E. Hirst, L. Goossens, A. George, and R. Weinkove, "Non-invasive detection of anaemia using digital photographs of the conjunctiva," *PloS one*, vol. 11, no. 4, p. e0153286, 2016.

[4] G. Dimauro, M. E. Griseta, M. G. Camporeale, F. Clemente, A. Guarini, and R. Maglietta, "An intelligent non-invasive system for automated diagnosis of anemia exploiting a novel dataset," *Artificial Intelligence in Medicine*, vol. 136, p. 102477, 2023.

[5] G. Dimauro, R. Maglietta, T. Bai, and S. Kasiviswanathan, "Eyes-defy-anemia," 2022. [Online]. Available: https://dx.doi.org/10.21227/t5s2-4j73

[6] P. Appiahene, K. Chaturvedi, J. W. Asare, E. T. Donkoh, and M. Prasad, "Cp-anemic: Aconjunctival pallor dataset and benchmark for anemia detection in children," *Medicine in Novel Technology and Devices*, vol. 18, p. 100244, 2023.

[7] P. Appiahene, E. J. Arthur, S. Korankye, S. Afrifa, J. W. Asare, and E. T. Donkoh, "Detection of anemia using conjunctiva images: A smartphone application approach," *Medicine in novel technology and devices*, vol. 18, p. 100237, 2023.

[8] M. Momtaz, M. N. Navin, R. Z. Rakin, K. Nur, and S. Parvin, "Accurate anemia disease detection using eye conjunctiva features with improved yolov9," in *2025 17th International Conference on Information Technology and Electrical Engineering (ICITEE)*. IEEE, 2025, pp. 1–6.

[9] O. Ramos-Soto, J. Ramos-Frutos, E. Perez-Zarate, D. Oliva, S. J. Mousavirad, and S. E. Balderas-Mata, "Non-invasive anemia detection from conjunctiva and sclera images using vision transformer with attention map explainability," *Scientific Reports*, vol. 15, no. 1, p. 44142, 2025.

[10] N. Sehar, N. Krishnamoorthi, and C. V. Kumar, "Deep learning model-based detection of anemia from conjunctiva images," *Healthcare Informatics Research*, vol. 31, no. 1, pp. 57–65, 2025.

[11] J. W. Asare, M. M. Ujakpa, S. Alornyo, E. A. Kyei, P. Modey, W. L. Brown-Acquaye, E. Freeman, and A. Coleman, "Anaemia detection in infants of 6–59 months using medical images: a multi-architecture convolutional neural network-based approach," *Journal of Electrical Systems and Information Technology*, vol. 12, no. 1, p. 44, 2025.

[12] dastaswr anewmia, "anemia detection dataset," https://universe.roboflow.com/dastaswr-anewmia/anemia-detection-yciqn-s7qxm, jun 2025, visited on 2026-07-14. [Online]. Available: https://universe.roboflow.com/dastaswr-anewmia/anemia-detection-yciqn-s7qxm

[13] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," https://github.com/ultralytics/ultralytics, 2023, version 8.0.0, AGPL-3.0 license.

[14] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2018, pp. 839–847.